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AI/ML Engineer | Forward Deployed Engineer

Senior AI Systems Architect and Forward Deployed Engineer with an established track record of designing, building, and deploying **23+ production systems** across the United States, United Kingdom, and Europe. Specializing in autonomous multi-agent pipelines, multi-cloud GenAI ops, vector databases, zero-trust cryptographic vaults, and peer-reviewed deep learning research.

🌐 Live Portfolio: ahsantariq.ranbval.com ↗ | 📁 Case Studies Hub (23 Projects) ↗ | 📧 ahsantariq0724@gmail.com | 🔗 LinkedIn Profile ↗ | 💻 GitHub: ahsantariq7 ↗

23+	$141M+	6	11,500+	81
PRODUCTION SYSTEMS	ACTIVE PIPELINE SYNC	PUBLISHED PAPERS	OPEN-SOURCE DOWNLOADS	LOGIC PUZZLES SOLVED

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EXECUTIVE OVERVIEW

What I Build & The Real-World Problems I Solve

A crystal-clear guide written in plain everyday English for founders, executives, non-technical managers, and recruiters.

💡 The Engineering Philosophy: Zero Manual Clutter, 100% Reliable Automation

Modern businesses lose hundreds of hours each month copying data between software, taking meeting notes by hand, chasing unpaid invoices, and fielding repetitive phone calls. I design and build automated software brains that take over repetitive office work, connect disconnected apps, and ensure systems run 24/7 with zero human errors and sub-second response times.

🤖 1. Autonomous AI Agents & Voice Specialists

I build AI phone agents and chatbots that speak naturally like human experts. They answer incoming customer calls in under 60 seconds, resolve client questions, check live calendars, book appointments on Zoom/Google Meet, and deliver concise action-item summaries directly into team Slack channels.

- < 600ms Voice Turnaround
- Zero Call Drops
- Automated Calendar Booking

⚡ 2. Business Automation & Multi-Ledger Accounting

I eliminate repetitive paperwork. My accounting engines read receipts and vendor bills automatically, verify they haven't been paid before, and sync them seamlessly across QuickBooks, Xero, Buildium, and NetSuite without humans having to type a single number.

- 75% Faster Closing
- Zero Double-Payments
- Multi-Currency Ready

🔒 3. Military-Grade Cloud & Cryptographic Security

I protect sensitive corporate data. Instead of keeping confidential passwords and keys in vulnerable text files, I engineer encrypted vaults with sub-2ms verification. My systems manage multi-cloud networks across AWS, Azure, and Google Cloud to automatically stop waste and lower server bills.

- NLnet EU Research Grant
- 11,500+ SDK Downloads
- Multi-Cloud FinOps

🔬 4. Deep Learning Research & Field Robotics

I conduct original peer-reviewed AI research and build self-navigating robotics. From autonomous 4x4 ground vehicles that map uncharted hazard areas using laser scans, to indoor drones with sub-5cm hovering accuracy and factory vision AI that spots product defects with zero prior training data.

- 6 Published Papers
- Sub-5cm Hover Accuracy
- 98.67% Cyber Threat Accuracy

GLOBAL FOOTPRINT

Trusted Across International Markets

🇺🇸 United States
Healthcare ERP, Real Estate Accounting, Legal Intake & FinTech

🇬🇧 United Kingdom
Multi-Ledger Tax Reconciliations & Corporate Portals

🇪🇺 European Union
NLnet NGI Zero-Trust Grant & Tourism Booking Engines

🇦🇺 Australia & PK
Multi-Jurisdictional Financial Vaults & AI Microservices

SECTION 01

Enterprise & Core Infrastructure

High-throughput financial engines, multi-tenant enterprise ERP portals, and cryptographic vaults

#01 Volvexer — Global Multi-Ledger Accounting & Tax Reconciliation

ENTERPRISE PLATFORM

Client / Org: Volvexer · Global Financial Services (USA, UK, Australia, Pakistan) **Timeline:** 2024 – Present

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Accountants spend dozens of hours every week re-typing numbers between QuickBooks, Xero, and bank feeds. Built an automated financial brain that syncs bank transactions, reads messy receipts with AI, and prepares tax reconciliation reports across US, UK, and Australian tax laws with zero human data entry.

- **Multi-Ledger Integration:** Bi-directional sync pipelines for QuickBooks, Xero, NetSuite, and Zoho Books with automated forex conversion.
- **Automated OCR Intake:** Machine learning document classifier splitting multi-currency invoices and bills, cutting month-end closing by 75%.

`Multi-Ledger Sync` `QuickBooks` `Xero` `NetSuite` `Zoho Books` `Python` `OCR` `Tax Vault`

Read Full Architectural Case Study: www.volvexer.com/ ↗

#02 Ambica International (Dovetail) — Healthcare Logistics & ERP Backbone

LIVE PRODUCTION

Client / Org: Ambica International · United States (Remote) **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Hospitals and pharmacies need urgent medical deliveries without delay. Transformed paper clipboard tracking into a 100% digital portal: live GPS tracking for 102 delivery trucks, instant sync with NetSuite for $141M in medicine orders, and 60-second digital contract approvals via DocuSign.

- **Sub-Second NetSuite Mirror:** 20-stream SuiteQL pipeline syncing 51,462 invoices and 35,935 purchase orders with live 3-second P&L ticking.
- **Zero-Trust Server Bridge:** 3-layer OTP authentication vault protecting on-premise physical databases with strict IP and attempt limits.

`GCP Cloud Run` `NetSuite SuiteQL` `Cartrack GPS` `DocuSign` `FastAPI` `PostgreSQL` `Redis`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ambica-dovetail/ ↗

#03 Ranbval Core & Secret Manager — Zero-Knowledge Cryptographic Vault

NLNET GRANT AWARD

Client / Org: Ranbval Security Platform · Global Enterprise SaaS **Timeline:** 2025 – Present

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Companies frequently leak sensitive passwords and API keys stored in plain text configuration files. Built an ultra-secure digital vault that encrypts secrets using military-grade cryptography, delivers keys in under 2 milliseconds, and shuts down instantly if tampering is detected. Awarded an EU NLnet Foundation grant.

- **Hardware-Grade Encryption:** AES-256-GCM and ChaCha20Poly1305 with offline verification and repo-bound cryptographic access tokens.
- **High-Adoption SDKs:** Authored open-source libraries on PyPI and npm with 11,500+ global downloads and sub-2ms pure ASGI decryption.

`AES-256-GCM` `ChaCha20Poly1305` `PyPI & npm SDKs` `Zero-Trust` `Schema Contracts`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ranbval-core/ ↗

SECTION 02

Autonomous Voice AI & Accounting Automation

Natural voice phone specialists, automated appointment booking, and multi-system invoice reconciliation

#04 Transact ULTD — Sub-Second Voice AI & Automated Merchant Engine

VOICE AI · LIVE

Client / Org: Transact ULTD · Financial Technology (USA) **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Credit card payment companies lose leads when callers wait on hold. Built a human-sounding AI voice agent that calls incoming merchant leads in under 60 seconds, calculates how much money they save on credit card processing fees, and transfers complex calls seamlessly to human specialists.

- **Sub-600ms Voice Turnaround:** Retell AI and LiveKit WebRTC audio pipeline maintaining natural conversational tempo without unnatural pauses.
- **Automated Lifecycle Engine:** Automatically parses merchant processing statements, creates CRM profiles, and triggers statement upload emails.

`Retell AI` `LiveKit WebRTC` `Twilio Trunking` `FastAPI` `HubSpot` `Python`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/transact-ultd/ ↗

#05 Terry Properties — AI Accounting & Multi-System Synchronization

PRODUCTION SYSTEM

Client / Org: Terry Real Estate Management · United States (Remote) **Timeline:** 2024 – 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Property managers were spending 25+ hours every week manually typing invoices from messy emails and WhatsApp photos into QuickBooks and Buildium. Built an AI accountant that reads any photo or PDF bill, checks that it hasn't been paid already, and posts it accurately across both accounting systems without human error.

- **Dual-System Ledger Mirror:** Real-time REST pipeline synchronizing invoices between Buildium and QuickBooks Online with conflict resolution.
- **Intelligent OCR & Deduplication:** Vision-based invoice parser extracting line items, tax IDs, and GL codes with 99.4% validation accuracy.

`GPT-4o Vision` `QuickBooks Online` `Buildium REST API` `Celery` `Redis`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/terry-accounting/ ↗

#06 Django Voice Agent — Phone AI & Autonomous Meeting Summarizer

AUTONOMOUS AGENT

Client / Org: Enterprise Sales Infrastructure · Commercial System **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Sales reps waste hours dialing cold leads and coordinating calendars. Built an autonomous AI phone assistant that calls prospects, talks naturally, checks live team availability, and books video calls on Zoom or Google Meet. When meetings finish, an AI bot delivers concise action items directly to Slack.

- **Full-Duplex Voice Pipeline:** Real-time streaming combining Deepgram Nova-2 speech-to-text, Claude reasoning, and ElevenLabs low-latency voice.
- **Autonomous Scheduling:** Resolves multi-attendee calendar conflicts, sends invites, and logs structured conversation summaries into CRM.

`Twilio SIP` `Deepgram Nova-2` `Claude 3.5 Sonnet` `ElevenLabs` `Google Calendar`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/django-voice-agent/ ↗

SECTION 03

Interactive AI, Digital Twins & Booking Engines

Real-time avatar synthesis, deterministic patient intake workflows, and conversational reservations

#07 Qira AI — Real-Time Avatar Intelligence & Social Digital Twins

CONSUMER AI PLATFORM

Client / Org: Qira AI · United States & Global **Timeline:** 2024 – 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Anyone can create a speaking 3D digital clone of themselves from just one selfie and voice sample. Friends can video call, chat, and play games with the digital twin, which talks authentically in their real voice and can even hang out with friends automatically when the real user is asleep or busy.

- **Sub-Second Video Synthesis:** Real-time lip-synchronization and facial animation streamed over WebRTC with sub-800ms round-trip latency.
- **Autonomous Persona Engine:** Memory-backed vector store preserving personality, conversational style, and user preferences across languages.

`WebRTC` `3D Avatar Rendering` `Lip-Sync Telemetry` `Multilingual STT` `Redis`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/qira-ai/ ↗

#08 Petroski Physiotherapy — Deterministic Marketing & Clinic Governance

HEALTHCARE AUTOMATION

Client / Org: Petroski Physio Clinic · Sports Rehabilitation (USA) **Timeline:** 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Sports therapists were constantly interrupted during patient appointments by marketers asking which athletes could be filmed for social media. Automated a polite Slack message to therapists 24 hours after a patient's evaluation; clicking a simple 'Yes/No' button updates marketing's board instantly without any office interruption.

- **Two-Board Deterministic Architecture:** Strict separation of clinical medical intake and marketing cards with zero PHI / HIPAA leakage.
- **Timed Slack Interactivity:** Automated 24-hour evaluation triggers with single-click interactive buttons updating Trello card state in real time.

`Slack API` `Trello REST API` `Webhook Router` `Python FastAPI` `Deterministic Logic`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/petroski-physio/ ↗

#09 Ballyhass Adventure Park — Autonomous Booking & Availability Engine

COMMERCIAL CHATBOT

Client / Org: Ballyhass Outdoor Adventure · Ireland / Europe **Timeline:** 2024 – 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Visitors to a busy outdoor adventure park struggled to find open time slots across different lakes and activities. Built a smart chatbot that understands everyday questions like 'Any spots next Friday for wakeboarding?', checks live tickets in 0.4 seconds, and lets families reserve instantly.

- **Live Availability Polling:** High-speed integration with FareHarbor booking API returning real-time ticket slots with sub-500ms response times.
- **Conversational Entity Extractor:** Translates conversational date/time requests into deterministic booking parameters with 1-click checkout.

`Voiceflow` `N8N Enterprise` `FareHarbor REST API` `Redis Cache` `PostgreSQL`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ballyhass-booking/ ↗

SECTION 04

Sales Intelligence & Cloud Automation Engines

Gemini context caching, self-hosted multi-CRM automation clouds, and film production telematics

#10 Prospect Intelligence Engine — Sales AI with Google Context Caching

ENTERPRISE AI

Client / Org: Enterprise Sales Intelligence · Contract Engagement **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Sales reps wasted 10 minutes digging through old call transcripts, LinkedIn posts, and client PDFs before answering buyer questions. Built an AI sales assistant that remembers the client's entire history using Google server caching, cutting AI bills by 90% and answering in less than 1 second.

- **Explicit Context Caching:** Gemini `/v1beta/cachedContents` integration caching 100,000+ tokens, reducing API costs by 90%.
- **4-Case Routing Engine:** Automated cache lifecycle manager handling creation, hydration, TTL expiration, and cold fallback.

`Google Gemini 2.5` `/cachedContents API` `N8N` `PostgreSQL` `Redis`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/prospect-intelligence/ ↗

#11 hddisenos — Self-Hosted Cloud Automation & Multi-CRM Synchronization

SELF-HOSTED CLOUD

Client / Org: hddisenos Creative & Industrial Agency · International **Timeline:** 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Commercial automation websites charge for every click and transfer, costing thousands every month. Built a private automation cloud on dedicated servers that runs 57 business operations—generating AI pet artwork, posting cars to Facebook for dealerships, and syncing customer records across HubSpot and Zoho.

- **Private Server Architecture:** Deployed containerized N8N and PostgreSQL on dedicated Coolify cloud, eliminating recurring SaaS fees.
- **Multi-CRM Webhook Hub:** Resilient two-way synchronization bridging HubSpot and regional Zoho CRM instances with zero data loss.

`Coolify Cloud` `N8N Dedicated` `HubSpot` `Zoho CRM` `PostgreSQL` `Google AI`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/hddisenos-automation/ ↗

#12 CoreImmersive — Hollywood Call Sheet Parsing & Set Telematics

MEDIA PRODUCTION AI

Client / Org: CoreImmersive Productions · Hollywood / Los Angeles (Remote) **Timeline:** 2024 – 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Hollywood movie sets operate off complicated 20-page daily schedules emailed late at night. Built an AI platform that reads messy schedule PDFs in seconds, organizes actor call times and shooting locations, and watches weather satellites to warn directors before rain hits outdoor movie sets.

- **Messy PDF Extraction:** Multi-column regex and LLM parsing pipeline extracting actor arrival times, locations, and equipment lists.
- **Predictive Weather Telematics:** Satellite radar monitoring issuing automatic alerts 90 minutes before rain hits outdoor filming sets.

`FastAPI` `PyPDF` `Claude 3.5 Sonnet` `OpenWeather Telematics` `Celery`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/coreimmersive-film/ ↗

SECTION 05

Scalable Platforms, Clinical Data & Legal AI

Microservice AI tool directories, cold-chain pharma data lakes, and multilingual immigration intake

#13 2456.ai — Autonomous AI Tool Discovery & Scalable Microservices

SAAS MICROSERVICES

Client / Org: 2456.ai · United States (Remote) **Timeline:** 2023 – 2024

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

With thousands of AI tools launched every month, businesses struggle to find what works. Built a smart platform that analyzes AI tools, tests their capabilities, and routes user questions to the best tool with sub-second response times and automated subscription billing.

- **Hybrid Microservice Architecture:** Django state management combined with decoupled FastAPI services for high-speed AI evaluations.
- **Vector Search & Stripe:** Pinecone vector embeddings for semantic tool recommendations and Stripe webhooks for tier-based billing.

`Django Core` `FastAPI Microservices` `Pinecone Vector DB` `PostgreSQL` `Stripe`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/2456-ai-platform/ ↗

#14 QPharma Global — Cloud Data Lake & Cold-Chain Telemetry Pipeline

ENTERPRISE DATA LAKE

Client / Org: QPharma Global · United States (Remote) **Timeline:** 2022

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Sensitive medicines spoil if they get too warm during international shipping. Built an automated cloud data pipeline that ingests millions of shipment temperature logs and sales records, instantly spotting damaged batches and cutting data reporting time by 70%.

- **Serverless ETL Engine:** AWS Glue and Lambda pipeline ingesting gigabytes of cold-chain telemetry and sales spreadsheets into S3.
- **Fast Analytical Querying:** Cleaned columnar data partitioned for Amazon Athena, enabling sub-second compliance and audit reporting.

`AWS Glue` `S3 Data Lake` `AWS Lambda` `Pandas` `Athena SQL` `CloudWatch`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/qpharma-data-platform/ ↗

#15 Sean Z & Associates — Multilingual AI Legal Intake Architecture

LEGAL AI ASSISTANT

Client / Org: Sean Z & Associates · Law Practice (USA) **Timeline:** 2024

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Immigration lawyers spend hours asking the same initial intake questions in English and Spanish. Built a bilingual AI legal assistant on WhatsApp and phone that interviews potential clients, extracts key case details, and prepares organized client files while keeping all data strictly private.

- **Bilingual Conversational Intake:** Real-time Spanish/English qualification extracting family history and visa eligibility facts.
- **Privilege-Bound Security:** Zero-retention AI prompts and encrypted storage ensuring strict attorney-client privilege compliance.

`Twilio Voice` `WhatsApp Business API` `OpenAI GPT-4` `Google Looker Studio`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/seanz-law-firm/ ↗

SECTION 06

Multi-Cloud Orchestration, Vector DBs & Super-Systems

Autonomous cloud cost optimizers, semantic RAG email engines, and 16-pillar collaborative ecosystems

#16 Jericho AI — Autonomous GenAI Ops & Multi-Cloud Orchestrator

MULTI-CLOUD PLATFORM

Client / Org: Jericho AI · Florida, United States **Timeline:** 2024 – 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Companies waste thousands of dollars leaving idle servers running across AWS, Google Cloud, and Azure. Built a smart multi-cloud platform where autonomous AI agents continuously monitor server health, detect wasted computing power, and optimize cloud workloads in real time without human delay.

- **Unified Cloud Abstraction:** Unified control plane orchestrating resources across AWS, Azure, GCP, IBM, and Oracle Cloud (OCI).
- **Multi-Agent FinOps Swarm:** Coordinated AI agent system predicting cost waste and executing automated scaling without service downtime.

`AWS` `Azure` `GCP` `IBM Cloud` `OCI` `Autonomous Multi-Agent Swarm` `FinOps`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/jericho-ai-multicloud/ ↗

#17 InstaMailAI — Scalable Vector Database & Semantic RAG Email Engine

VECTOR SEARCH RAG

Client / Org: InstaMailAI · United States (Remote) **Timeline:** 2025

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Traditional email search only works if you remember the exact words used months ago. Built a smart vector search brain that understands the underlying meaning of emails, allowing business teams to find past discussions instantly and draft perfectly contextual replies.

- **Dense Vector Search Pipeline:** MongoDB Atlas Vector Search indexing high-dimensional email embeddings with sub-50ms query latency.
- **Real-Time RAG Synthesis:** Context-grounded draft generator reducing email response time by 80% with zero hallucination.

`MongoDB Atlas Vector Search` `OpenAI Embeddings` `LangChain` `FastAPI` `HNSW Index`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/instamail-ai/ ↗

#18 Ranbval Social Super-System — 16-Pillar Collaborative Ecosystem

PROPRIETARY ECOSYSTEM

Client / Org: Ranbval Platform · Lead Architect **Timeline:** 2024 – Present

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Modern teams struggle using 10 different apps for chat, tasks, video calls, and files. Built a unified platform that combines encrypted real-time chat, automated AI assistants, private file vaults, and team collaboration into one seamless application running under a single secure web address.

- **Unified Single-Root Platform:** 16 integrated modules sharing a unified authentication session and cryptographic access token layer.
- **Real-Time WebSocket Fabric:** Sub-50ms live chat, presence telemetry, and ephemeral conference calls powered by Redis Sentinel.

`FastAPI` `WebSocket` `React` `WebRTC` `Redis Sentinel` `Single Secure Root`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ranbval-social/ ↗

SECTION 07

Peer-Reviewed AI Research & Cybersecurity

Zero-shot manufacturing defect benchmarks, poisoning-resistant federated learning, and high-speed NIDS

#19 Industrial Defect Localization — Zero-Shot vs Few-Shot Benchmark

ACADEMIC RESEARCH

Client / Org: Computer Vision & Industrial AI Research · Sole Author: Ahsan Tariq **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Factory assembly lines lose money when cameras miss defective parts or when AI requires thousands of defect photos to learn. Evaluated 3 modern AI vision paradigms across 15 manufacturing categories (5,354 images), proving which model spots microscopic flaws with zero prior training data.

- **Comprehensive 15-Category Tour:** Benchmark across 10 rigid objects and 5 non-rigid textures measuring Image-AUC and Pixel-PRO.
- **Empirical Anomaly Map Analysis:** Validated boundary localization trade-offs between vision-language models and foundation features.

`MVTec AD Dataset` `CLIP` `DINOv2` `Segment Anything (SAM)` `PRO-AUC` `PyTorch`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/mvtec-defect-benchmark/ ↗

#20 Robust Federated Learning — Defense Against Edge Model Poisoning

ACADEMIC RESEARCH

Client / Org: Distributed AI & Privacy-Preserving Research · Sole Author: Ahsan Tariq **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Hospitals or self-driving car fleets can train a shared AI together without sharing private patient or location data. Built a mathematical trust guard that inspects updates from every computer, instantly spotting and kicking out hacked or malicious nodes trying to poison the AI.

- **Trust as a Gate Protocol:** Binary admission filter gating model weights based on historical angle and gradient alignment.
- **99.1% Poisoning Mitigation:** Maintained high perceptual accuracy under adversarial Byzantine attacks where standard FedAvg failed.

`Trust-Gated Aggregation` `Cosine Similarity` `Byzantine Defense` `Edge Networks`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/robust-federated-perception/ ↗

#21 High-Throughput Network Intrusion Detection — ANOVA F-Test NIDS

ACADEMIC RESEARCH

Client / Org: Cybersecurity & Machine Learning Research · Sole Author: Ahsan Tariq **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Modern network firewalls get overwhelmed trying to analyze dozens of technical parameters for every packet. Engineered a high-speed cybersecurity pipeline using statistical math to eliminate 70% of network clutter, detecting cyberattacks with 98.67% accuracy in less than a millisecond.

- **69.4% Dimensionality Reduction:** Selected 15 top-ranked ANOVA flow features from 49 raw attributes without losing attack signal.
- **State-of-the-Art Speed:** 98.67% binary detection accuracy via variance-smoothed Gaussian Naive Bayes, outperforming complex neural models.

`UNSW-NB15 Dataset` `ANOVA F-Test` `Gaussian Naive Bayes` `XGBoost` `98.67% Acc`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/unsw-intrusion-detection/ ↗

SECTION 08

Field Robotics & Autonomous UAV Flight Control

Rugged ground robot SLAM navigation, cascade PID drone flight dynamics, and smart wheelchair autonomy

#22 Clearpath Husky A200 UGV — Autonomous SLAM & Dijkstra Navigation

FIELD ROBOTICS

Client / Org: Field Robotics & Autonomous Ground Systems · Clearpath Robotics **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Built an autonomous brain for a heavy-duty 4x4 outdoor robot (Clearpath Husky). The robot navigates through unmapped hazardous environments, creates a digital 2D map in real time using laser scans, and drives safely to mission waypoints without human remote control.

- **20mm Occupancy Grid SLAM:** Rao-Blackwellized particle filter SLAM with calibrated differential skid-steer odometry (wheel base 1.07m).
- **Dynamic Obstacle Avoidance:** Dual-layer costmap inflation combined with Dijkstra global planning and Dynamic Window local control.

`ROS Navigation Stack` `Gmapping SLAM` `Dijkstra Planner` `DWA Local` `Odometry` `Gazebo`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/husky-slam-navigation/ ↗

#23 Hector Quadrotor UAV & Smart Wheelchair — Flight Dynamics & Autonav

AERIAL & GROUND ROBOTICS

Client / Org: Aerial Robotics & Autonomous Assistive Mobility · ROS & Gazebo 3D **Timeline:** 2025 – 2026

IN PLAIN ENGLISH: PROBLEM SOLVED & BUSINESS RESULTS

Engineered two high-precision autonomous robotics systems: 1) An autonomous quadrotor drone that takes off, hovers steadily in mid-air with sub-5cm altitude error, and follows complex 3D waypoints. 2) A self-driving smart wheelchair that maneuvers through narrow hospital doorways and corridors with zero collisions.

- **100 Hz Cascade Flight Controller:** Inner-loop attitude rate PID coupled with outer-loop position guidance and automatic motor arming guard.
- **Assistive Corridor Trajectory:** Dynamic laser scan envelope maintaining safe wall clearance in constrained indoor environments.

`Hector Quadrotor` `Cascade PID Flight Controller` `Gazebo 6-DOF` `Smart Wheelchair Autonav`

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/hector-quadrotor-uav/ ↗

PROPRIETARY PRODUCTS

SaaS Platforms I Architected, Built & Own

Beyond client engagements, I design and operate proprietary software ecosystems built for extreme scale, security, and financial reliability.

1. Ranbval — The 16-Pillar Collaborative Platform

LEAD ARCHITECT · LIVE

Ranbval (ranbval.com) is an all-in-one collaborative ecosystem engineered under a single secure root. It replaces dozens of disconnected subscription tools with unified team chat, AI digital assistants, encrypted file vaults, dynamic task boards, and social connection spaces—all protected by zero-trust cryptographic access.

Vault-First Architecture
Zero-knowledge cryptographic access ensuring user passwords and files are inaccessible to unauthorized parties.

Sub-50ms WebSocket Mesh
Live chat, real-time presence indicators, and instant group notifications powered by Redis Sentinel.

16 Unified Applications
Chat, Video Conferencing, File Vault, Task Management, AI Workspaces, and Social Feed under 1 login.

Read Ranbval Platform Case Study ↗ Read Ranbval Security Case Study ↗

2. Volvexer — Automated Global Multi-Ledger Accounting

LEAD SYSTEMS ARCHITECT

Volvexer (volvexer.com) is an automated accounting service engineered for multinational businesses operating across the United States, United Kingdom, Australia, and Pakistan. It bridges disparate ledgers, automates monthly tax reconciliations, and runs high-accuracy OCR invoice extraction.

4 Global Jurisdictions
Pre-configured tax and GAAP/IFRS reconciliation compliance rules for US, UK, AUS, and Pakistani law.

Bi-Directional ERP Sync
Connects live with QuickBooks Online, Xero, NetSuite, and Zoho Books with automated forex conversion.

Zero-Leak Document Vault
Encrypted multi-tenant document repository ensuring zero audit discrepancies and SOC 2 readiness.

Visit Volvexer Platform: volvexer.com ↗

3. Open-Source Ecosystem & Package Footprint

GLOBAL DEVELOPER TOOLING

Active creator and maintainer of developer libraries on PyPI (Python) and npm (Node.js) with **11,500+ global downloads**. Tools include cryptographic secrets verification, high-performance ASGI decryptors, and automated context-caching connectors.

PyPI: ranbval-client
Pure ASGI automatic environment secret decryption library for high-speed FastAPI runtimes.

npm: @ranbval/vault
Production Node.js / TypeScript client for zero-latency cryptographic token verification.

GitHub Repositories
Verified open-source machine learning pipelines, NIDS algorithms, and ROS navigation stacks.

ACADEMIC & PROFESSIONAL VALIDATION

Scientific Publications & Peer Reviewer Appointments

Demonstrated peer-reviewed contributions to machine learning, cybersecurity, computer vision, and autonomous robotics.

1. Three Paradigms, One Benchmark: A Diagnostic Grand Tour of Zero-Shot vs Few-Shot Anomaly Localization on MVTec AD

Sole Author: **Ahsan Tariq** · Journal of Independent Studies and Research - Computing (JISR-C) · 2026

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3. ANOVA F-Test Feature Selection for Machine Learning-Based Network Intrusion Detection on the UNSW-NB15 Dataset

Sole Author: **Ahsan Tariq** · Journal of Independent Studies and Research - Computing (JISR-C) · 2026

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4. Deep Learning Architectures for Robust Medical Imaging Diagnostic Localization

Lead Author: **Ahsan Tariq** · Peer-Reviewed Medical AI Research · 2024

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5. Autonomous Mobile Robot Navigation and Dynamic Obstacle Avoidance via Deep Reinforcement Learning

Author: **Ahsan Tariq** · Robotics & Autonomous Systems Proceedings · 2024

Verified Robotics Research

6. Cryptographic Access Control and Zero-Trust Secrets Distribution for Microservice Swarms

Author: **Ahsan Tariq** · Security & Distributed Systems Technical Report · 2025

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